Quantum Dice You Can Audit

Team: **Bell & Order**
Alvaro Quinones Jauregui Damien Koon
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Abstract

Write at the end

Preamble

[DK: Hi Alvaro! I wrote up a little bit to get started. Let me know if you have questions.] You can use [AQJ: This command] to write comments. You can also use **[DK: This command]** to make bolded comments. Just make sure to use your initials when writing ofc. You can change the color above by the command definecolor

Damien todo:

- Skim Pironio 2010 and Kavuri 2025
- Expand slides
- Finish analysis as to why this is a good method
- Determine broader implications of this system
- Recruit a judge to play the cheater
- **After some results:**
- Write results section, prepare final presentation
- Review abs, concl, intro, etc

Alvaro todo: [DK: I just added stuff, do as you please, don't feel like you have to stick to this]

- Get the CHSH code running on the current version of Qiskit
- Note: Use Sampler and not Estimator (this saves the shot's bits to determine the winner)
- Run first job on the simulator
- Run first job on quantum hardware
- Make a program to compute S and its error, flag anyone whose S isn't above 2 by at least 5σ.
- Write the cheater script to give one of the judges
- Combine remaining honest players' bits to reliably, fairly, and randomly pick a winner
- Upload all code to GitHub and the Hackathon page

1 Introduction

Problem Statement: Public random draws, such as raffles and jury selection, require trust in whoever runs them. A classical random number generator can be exploited since we cannot prove it was not seeded, scripted, or tampered with. We want to use quantum computing to make random draws independently checkable.

Goal: Build a drawing system where the random bits come from a Bell test on IBM quantum hardware, and anyone can check three things after the drawing:

- The list was fixed before the draw
- The bits used to determine the draw came from published IBM jobs
- A measured value certifies how unpredictable the bits were, and reveals who tampered with them.

2 Who this Affects

This can affect raffle organizers, participants, jury pools, lotteries, and more. As entrants enter a drawing, they are forced to trust the operator. With this system, we can remove that trust in the organizer leading to more reliable jury pools and fair lotteries.

3 Background: The CHSH Test

3.1 Entanglement

Two qubits starting in $|00\rangle$ are entangled by a Hadamard gate followed by a CNOT, producing the Bell state

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle + |11\rangle\right]$$

Each qubit alone gives 0 or 1 with equal probability, but the two outcomes are perfectly correlated. Let's call our two qubits Alice and Bob.

3.2 The CHSH Parameter

Alice measures at one of two angles, A_0 or A_1, and Bob measures at one of two angles, B_0 or B_1. Each of these measurements will give either $+1$ or -1. For each choice of angles, we can compute the correlation by

$$E(A_i, B_j) = \frac{N_{\text{same}} - N_{\text{different}}}{N_{\text{total}}}$$

Combining the 4 possible combinations into one parameter gives the CHSH parameter [2].

$$S = E(A_0, B_0) - E(A_0, B_1) + E(A_1, B_0) + E(A_1, B_1)$$

The minus sign on the 2nd term is a matter of convention, but at least one term must be subtracted. Without it, predetermined answers could make every term +1, giving $S = 4$, and no separation between classical and quantum would be possible.

3.3 The Classical Bound

Suppose every outcome were decided ahead of time by some hidden instructions. To denote this, we switch from A_i, B_j to a_i, b_j. Alice's answers a_0, a_1 and Bob's answers b_0, b_1 are fixed values of ± 1. Then we can factor this as

$$S = a_0 [b_0 - b_1] + a_1 [b_0 + b_1]$$

Since $b_0 = \pm 1, b_1 = \pm 1$, one bracket will always be zero and the other will be ± 2, therefore no matter Alice's choice of a_0, a_1,

$$|S| \leq 2$$

Any process where the outcomes are predetermined will satisfy this bound [1]. Looking ahead, we will see that quantum effects produce $|S| > 2$. Therefore, if this bound is not violated, it would indicate tampering.

3.4 The Quantum Value

Start with

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Apply the measurement angles through $R_Y(-\theta)$, which gives the correlation to be

$$E = \cos(\theta_A - \theta_B)$$

Choose

$$A_0 = \tfrac{\pi}{2}, \quad A_1 = 0, \quad B_0 = \tfrac{\pi}{4}, \quad B_1 = -\tfrac{\pi}{4}$$

This gives three correlations of $\frac{\sqrt{2}}{2}$ and one of $-\frac{\sqrt{2}}{2}$. Plugging these into S gives

$$S = 2\sqrt{2} \approx 2.828$$

This is the largest value quantum mechanics allows, known as Tsirelson's bound [3]. A more complete derivation and a previous hardware run on `ibm_kyiv` are given in [6].

4 Why Use Quantum Computing To Solve This Problem

Measurements on an entangled pair of qubits that violate the CHSH bound $|S| \leq 2$ cannot have been decided ahead of time by any hidden instructions, so no one, including the operator, could have known the outcomes in advance. The further S is above 2, up to $2\sqrt{2} \approx 2.828$, the less predictable the bits are. If someone faked or scripted the bits, they would not produce a violation, so a low S exposes tampering.

Using a Bell violation to certify randomness was first demonstrated by Pironio et al. [4], and NIST now runs a public randomness beacon built on a loophole-free Bell test [5]. Those systems certify a single trusted source. Our contribution is a draw in which each player supplies their own certified share, so the test not only detects tampering but also identifies who tampered.

[DK: Deeper analysis on why this is a better method than other methods both in qc and non-qc approachs]

5 Protocol

Each player runs their own CHSH job privately and obtains a bit string and a value of S. If a player is flagged as cheating, their bit strings are removed. The final draw only uses bit strings that pass the test.

- Publish the list of entrants. Each entrant is indexed from 0 to $n - 1$.
- Each player privately runs the circuit using Qiskit [7] and publishes a hash of their bit string.
- Everyone then reveals their bit strings and job IDs, which anyone can check against the published hashes and IBM's records.
- Compute S for each player. Any player whose S is not above 2 by at least 5σ is flagged as a cheater and has their bits removed.
- The remaining bits combine to pick the winner.

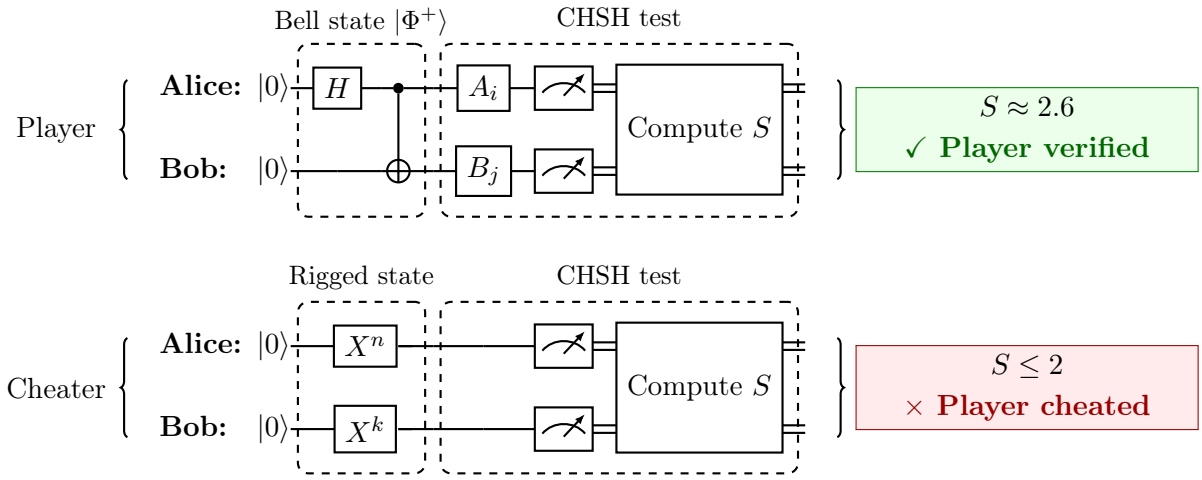


Figure 1: **Top:** An honest player entangles their qubits and measures in settings A_i, B_j. It is computed that $S \approx 2.6$[DK: This 2.6 is kind of arbitrary. We can use a real measured value once we have a hardware run.], violating the classical bound. The bits could not have been decided in advance.
Bottom: A cheater skips entanglement and rigs the qubits to predetermine the winner. By fixing the outcome, S is computed to be at most 2. This doesn't violate the classical bound and flags the cheater.

5.1 Picking a Winner

Each entrant gets a number. We read the combined bits as a number, and whoever holds that number wins. If it does not match anyone, we read the next few bits instead.

[DK: This can be made more specific later]

6 Limitations

While this system can be used to audit raffles, it has multiple limitations to be aware of:

- In our demonstration, both qubits sit on the same chip, so this is not a loophole-free Bell test. A device could be built to fake the correlations and pass the test.[DK: Cite/maybe define loophole-free Bell test earlier]
- Verifiers must still trust IBM's record of each job. An advanced bad actor could alter IBM's record to bypass the Bell test.
- Hardware noise can lower S, so an honest player may be wrongly flagged. Further redundancy is needed to avoid false accusations.
- The IBM Open Plan allows 10 minutes of QPU time per 28 days, which limits how many players can join one draw for one account.

7 Conclusions

For a robust system, we would separate qubits across chips so that each player can create loophole-free Bell tests. To eliminate a potential attack vector, we can eliminate trust in IBM and use a robust decentralized ledger. This is no small feat and introduces other points of attack, but it reduces overall risk. We also want to trigger a retest for S and ensure low noise before unfairly accusing a player.

References

[1] J. S. Bell, "On the Einstein Podolsky Rosen Paradox," *Physics Physique Fizika* **1**, 195 (1964).

[2] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed Experiment to Test Local Hidden-Variable Theories," *Phys. Rev. Lett.* **23**, 880 (1969).

[3] B. S. Tsirelson, "Quantum Generalizations of Bell's Inequality," *Lett. Math. Phys.* **4**, 93 (1980).

[4] S. Pironio *et al.*, "Random numbers certified by Bell's theorem," *Nature* **464**, 1021 (2010).

[5] G. A. Kavuri *et al.*, "Traceable random numbers from a non-local quantum advantage," *Nature* (2025), arXiv:2411.05247.

[6] D. Koon, "Violating the CHSH Inequality: Empirical Evidence of Quantum Nonlocality" (2025), `https://github.com/acroscopic/CHSH`.

[7] Qiskit contributors, *Qiskit: An Open-Source Framework for Quantum Computing* (2025), `https://github.com/Qiskit/qiskit`.